9 Designing for maintenance and the cleaner environment

9.1 – Characteristics, application, advantages and disadvantages of ‘cleaner’ design and technology – a product’s life cycle in relation to the following sustainable development issues

Topic	What students need to learn:
9 **Designing for maintenance and the cleaner environment**	9.1 – Characteristics, application, advantages and disadvantages of 'cleaner' design and technology – a product's life cycle in relation to the following sustainable development issues: *a) material selection – source, quantity, quality, range, recyclability, biodegradability* *b) manufacture – minimising energy use, simplification of processes, achieving optimum use of materials and components, giving consideration to material form, cost and scale of production*

Topic	What students need to learn:
9 Designing for maintenance and the cleaner environment	*c) distribution – efficient use of packaging, reduction of transport, alternatives to fossil fuels* *d) use – repair versus replacement, energy efficiency, efficiency ratings* *e) repair and maintenance –standardisation, modular construction, bought in parts* *f) end of life – design for disassembly, recovered material collection, sorting and re-processing methods, energy recovery, environmental implications of disposal to landfill.*

‘Cleaner’ design and technology…

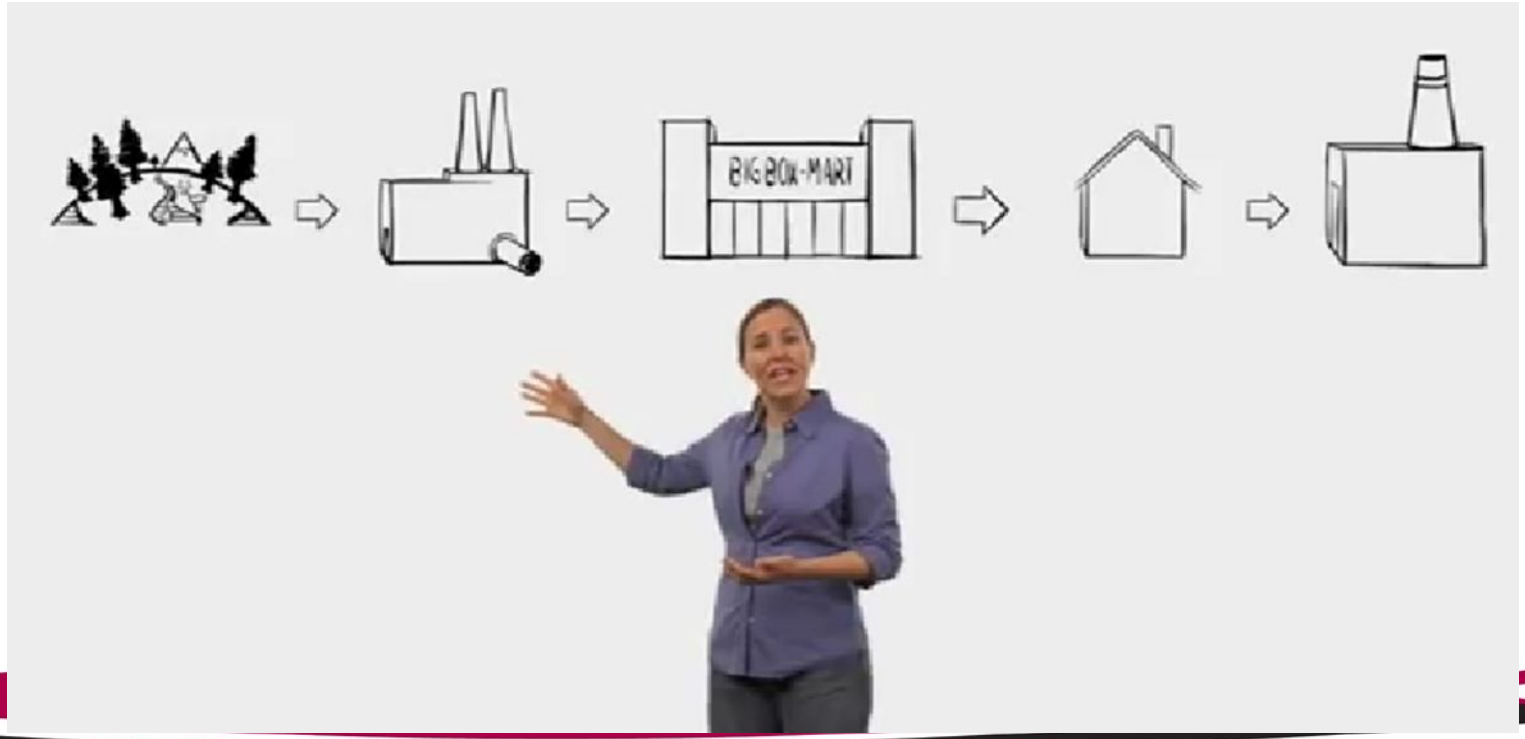
- Society is increasingly aware of and concerned about issues relating to the sustainability of products throughout their entire life cycle.
- This means that designers and manufacturers need to seek more sustainable ways of providing for people’s needs.
- Sustainability means the management of resources to minimise depletion and pollution. There are ‘six’ principle ways to achieve this.

The ‘Six’ R’s of Sustainability…



Product Life Cycle Diagram…

Material Extraction/Selection	Manufacture	Distribution	Use	End of Life



Material Selection…

Material Selection…

- The choice of material affects the use, care and disposal of products.
- This can be seen clearly when you consider an example such as the manufacture of drinking vessels for different purposes, for example *crystal glass and bone china at one extreme to melamine and polystyrene at the other*.
- See your notes on materials.

Material Selection…

- Labelling of materials to aid separation for recycling is achieved by the use of a standardised system of codes, usually moulded into the product.

Definition:

- **Recycling** - recovering the parts and materials from unwanted products to be used again.

Sources…

- The extraction and processing of large quantities of raw materials is often damaging to the environment and consumes a large amount of energy.
- Crude oil, required to manufacture polymers, is a finite resource that can additionally cause environmental problems during extraction and transportation.

Sources…

- Higher rates of recycling and seeking alternative materials are, therefore, high priority targets.
- The production of raw materials is highly energy dependent, particularly for processes such as smelting aluminium.
- This process requires 13 kilowatt hours (46.8 megajoules) electrical energy to produce one kilogram of aluminium.

Sources…

- High levels of emissions such as CO_2 are also an unfortunate by-product of such processes.
- Tragic flooding events at the ***Minas Gerais iron ore mine in Brazil*** and the ***methyl isocyanate gas escape disaster at a chemical plant in Bhopal in India*** demonstrate the potential dangers to life and the environment of some raw material extraction and manufacturing processes.

Sources…

- Timber and other natural materials offer potentially friendly alternative materials for some products, but deforestation and environmental degradation can result.
- The Forest Stewardship Council (FSC) scheme promotes the sustainable management of these resources.

Quantity…

Reducing the quantity of materials used in products is very important. This can be achieved by strategies such as:

- Heinz has saved 1400 tonnes of steel a year, without sacrificing quality, by ***reducing*** the weight of the ***'easy open ends'*** of its cans.
- In furniture manufacture, dedicated software can be used to plan how multiple parts can be nested in a piece of sheet material, to minimise unusable off-cuts.

Quantity…

- The extensive use of plastics in toy manufacturing is currently under close scrutiny, but rotational or blow-moulded, one-piece, hollow mouldings are inherently strong and use the minimum possible amount of material.

Quantity…

Definition:

- **Reduce** – minimising wastage and excess materials or packaging.

Recyclability…

- The ‘circular economy’ is a way of organising materials supply, manufacture and product end-of-life in the most environmentally friendly way possible.
- It is increasingly important in conserving dwindling resources and maintaining high quality materials through the facilitation of more effective recycling.
- **Upcycling** – is becoming a fashionable movement, whereby unwanted products and materials are **reused** creatively.

Recyclability…

- Examples include furniture from old pallets, clothing and accessories made from discarded festival tents and bags and belts made from discarded tyre inner tubes.

Biodegradability…

- **Biodegradable** materials decompose in a natural environment, which is important in order to minimise pollution.
- Degradation of **biopolymers** occurs due to the action of micro-organisms, and the process produces water, carbon-dioxide, biomass and methane.
- A range of natural and synthetic *'biopolymers' (biodegradable polymers)* has been developed to respond to this.

Biodegradability…

Biopolymer Examples:

- **Cornstarch polymer:** this is a natural biopolymer made from vegetable's such as potatoes, corn and maize that have high starch content. It is used to manufacture disposable cups, cutlery and takeaway packaging.
- **Polylactide acid (PLA):** a synthesized lactic acid, derived from fermented sugar cane or corn kernels, creates polylactide acid which is used for bottles, nappies, medical products and 3D printing.

Biodegradability…

Biopolymer Examples:

- **Lactide:** this is a water soluble and fully composable synthetic biopolymer that is used extensively in medical applications and sachets for detergents.
- **Polyhydroxybutyrate (PHB):** this is a bacteria fermented sugar and carbohydrate-based material which is commonly known as ***Biopol®***. It is used for packaging, although it isn't resistant to chemicals such as bleach and has a lower impact resistance than many other polymers.

Biodegradability…

Biopolymer Examples:

- **Bio-hatch:** isn't a biopolymer as such but an additive, which at concentrations of about 1% will render a range of conventional plastics such as HDPE and PP completely biodegradable within 1 – 5 years.

Biodegradability…

Biodegradable polymers do have some **disadvantages:**

- Degradation of some types *releases potentially harmful* ***methane gas*** from landfills.
- Recycled conventional polymers ***'contaminated'*** by biodegradable polymers suffer from a reduction in value and some of their properties are adversely affected, which may result in lower recycling rates.

Biodegradability…

Biodegradable polymers do have some **disadvantages:**

- They tend to be more expensive than conventional polymers as their production doesn't yet benefit from comparable economies of scale.
- Procedures for disposal must be adhered to and some have very specific requirements. An example of this is PLA, which needs a high temperature, oxygen and other special conditions only found in industrial composting facilities in order to biodegrade.

Biodegradability…

Biodegradable polymers do have some **disadvantages:**

- Growing the raw materials requires land that would otherwise be used for vital crops.
- The cold temperatures found in oceans do not facilitate decomposition, which may result in harmful floating plastic or micro-plastic particles.

Recyclability…

Definition:

- **Upcycling** – creative reuse of unwanted products and materials.
- **Reuse** – making it possible to use products or parts again.
- **Biodegrade** – the decomposition of materials in a natural environment.
- **Biopolymer** – a biodegradable polymer.

Manufacture…

Manufacture…

Strategies that allow manufacturer's to minimise their energy use and optimise their use of materials and components:

- Energy audits are carried out by specialists to establish potential savings. In-house, cross-department energy management teams can be organised, possibly accompanied by bonus incentives linked to energy savings.

Manufacture…

- Adopting ISO 50001 (Energy Management) guidelines to reduce energy use and emissions.
- Simplified designs with fewer components, lower material costs and quicker assembly times.
- Consideration of adopting a single stage manufacturing process for components, such as ‘die-casting’, to replace a multi-stage series of wastage and fabrication procedures.

Manufacture…

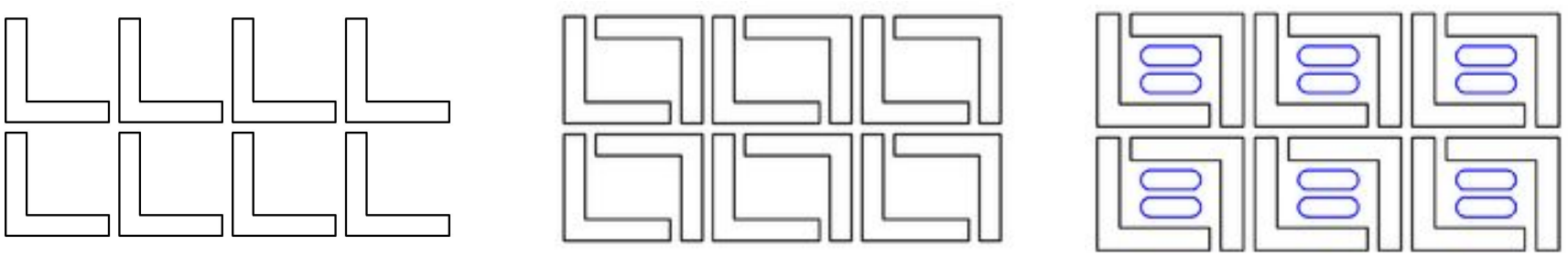
- Consideration of alternative materials with lower energy requirements for processing and those that produce less waste.
- Reorganisation of manufacturing to optimise workflow and quality.
- Careful organisation of when machines are used can facilitate the optimisation of energy use, particularly if they can be operated outside of peak hours.

Manufacture…

- Installation of energy saving technologies such as automatic lighting and heat recovery systems, as well as the employment of alternative energy sources such as solar panels and wind turbines.
- The incorporation of lightweight, honeycomb structures in products, such as some Ikea furniture, which use far less material than conventional boards.

Manufacture…

- The use of alternatives such as ‘glulam’ or ‘fibre composites’ rather than traditional materials.
- Nesting sheet material components as efficiently as possible for cutting processes.



Manufacture…

- Installation of dedicated machinery to replace CNC machines when the scale of production rises to a sufficiently high level.
- Use stock sizes of materials and standard forms to facilitate easier sourcing and processing.

Distribution…

Distribution…

- The efficiency of distribution is helped by **Electronic Point of Sale (EPOS)**, which is a computer-based scanned barcode system for capturing data when consumers purchase goods.
- Efficient packaging involves ensuring that space in vehicles is not wasted. Examples include:
  - *Tetra Classic® Aseptic Cube food packages, which tessellate in 3D to form a cube when six of them are placed together.*

Tetra Classic® Aseptic Cube Food Packages…

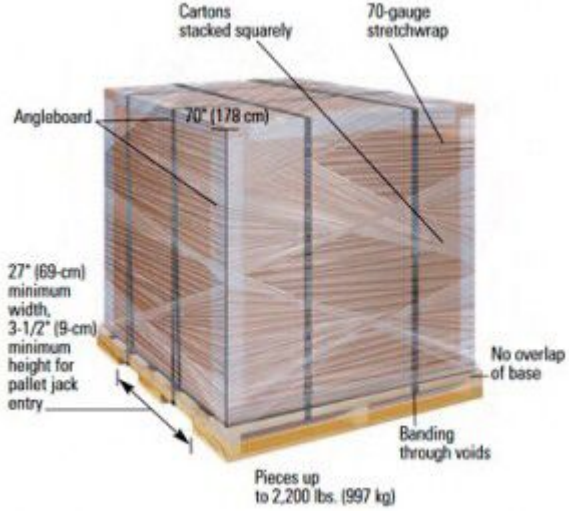


Distribution…

- *Self-assembly furniture made by companies such as Ikea, which is distributed in flat pack form.*



- *Standardised carton and pallet sizes that are compatible with freight containers, so there is no wasted space.*



- *Increased use of concentrates and refills, as well as packaging materials that are lighter, recyclable and made with recycled materials.*



Distribution…

- Production, planning and control (PPC) computer systems play an important part in ensuring that distribution is carried out efficiently.
- Companies are making greater use of central distribution centres, as these are more cost-effective and better in terms of environmental impact than dispersed systems.



Distribution…

- Initiatives such as the ***Voluntary Logistics Emissions Reduction Scheme (LERS)*** for the UK road freight industry, record, report and reduce transport emissions.

Distribution…

The transportation of goods can make use of alternative fuels and reduce environmental impact in a number of ways:

- Increasing the use of rail transport for freight makes use of electricity that may be from solar, hydro and wind sources.
- Initiatives such as ***Freight Electric Vehicles in Urban Europe (FREVUE)*** are exploring the potential viability of the greater use of electric vehicles.

Distribution…

- Some container ships are now starting to use ***Liquefied Natural Gas (LNG),*** which is far less polluting than traditional fuel. Even lower, and some potentially zero emission solutions for ships, such as hydrogen, methanol, ammonia and battery propulsion, still require a great deal of research and investment before they are likely to be viable.
- Maximise use of local suppliers, particularly in connection with JIT production.

Use…

Use…

- The near impossibility of gaining access to the internal components of a variety of products makes it very difficult for them to be serviced or ***repaired***.
- This ***built-in obsolescence*** results in many products, particularly electronics, being prematurely discarded.
- This benefits the company selling the replacement products but creates an enormous environmental problem.

Use…

- The energy efficiency of products is an important factor that should be borne in mind by consumers.
  - *The benefits of choosing efficient products include the minimisation of their own energy bills as well as reduced environmental damage.*
  - *Energy efficiency ratings are shown in the **EC Energy Label** attached to the electrical appliance.*
  - *This is a compulsory European scheme to indicate the energy consumption of many appliances.*

Use…

- The label must be clearly displayed for viewing by customers.
- A highly visible, coloured, A+++ to G scale facilitates ease of comparison.

Repair and Maintenance

Standardisation…

- Using standardised parts makes it easier to use bought-in components to manufacture and repair products.
- Manufacturers can be sure of parts' interchangeability, particularly if they are accredited by an independent body such as BSI.
- Examples include nuts and bolts, fuses, inkjet cartridges, plumbing fittings and lock barrels.

Maintenance…

- Designers and manufacturers need to consider how their products will be kept in a serviceable condition.

Relevant questions designers might ask…

- *Does the designer wish to make maintenance possible and who will carry it out?*
- *Are difficult to maintain watertight seals and other protective elements required?*

Maintenance…

- *Should non-specialists be deterred from making repairs, due to complex technology? For example, should specialised 'dealer only' tools and diagnostic equipment be required for car repairs?*
- *Should adhesive joints, facilitating slim smartphones and tablets, be used as they make access difficult?*

Maintenance…

- *Should temporary and integral fixings be used?*
  - *Temporary joining methods such as screws for access panels encourage home repair.*
  - *Integral, tool-free fittings facilitate regular maintenance such as battery replacement.*

Modular Construction…

- Modular design and construction is the use of interchangeable units, sharing common interfaces within a designed product or system, which facilitates easier upgrades and repairs.
- Computers are good examples of this and the standardised interfaces that are generally used for hard drives, power supplies and processors facilitate this.

Modular Construction…

- Temporary buildings are made using a modular design so that they can be assembled in a variety of easily joined units and can easily be replaced or repaired.
- Kitchen units and cameras / lens systems are other examples of modular systems making use of standardised sizes and interfaces to facilitate upgrading, replacement and repair.

Modular Construction…

- Consumers usually replace mobile phones as newer models become available.
- **'Fairphone'** (*https://www.fairphone.com/en/*) is an example of an open source software-based, modular, easily upgraded and repaired mobile phone challenging the current trend to discard mobile phones every couple of years.

Exam Tip

Don't concentrate solely on ***recycling*** when you are asked to explain how products can be made more ***'sustainable'***.

Try to include a ***range*** of issues, such as *choice of materials, efficiency of distribution and potential for making the designs modular to facilitate easier repair or upgrades.*

End of Life…

Disassembly…

- Products designed for ease of ***disassembly*** at the end of their useful life are much less damaging to the environment.
- Being taken apart easily into their various components and materials makes them less likely to end up as ***landfill***. Serviceable parts may be reused, or the materials can be fed back into the supply chain; this is referred to as a ***'circular economy'***.

Disassembly…

- This is very important for non-renewable materials such as metals, glass and most polymers.
- Ease of disassembly is critical to the design of products such as cars and white goods.
  - *Legislative directives such as End of Life Vehicles (ELV) and Waste from Electrical and Electronic Equipment (WEEE) have encouraged ease of disassembly.*
  - *Cars are difficult to recycle due to dealing with depollution, automobile shredder residue (ASR) and, increasingly, lithium batteries in electric and hybrid vehicles.*

Disassembly…

- ***'Active disassembly'*** involves products like mobile phones automatically separating into their components parts in specially designed recycling processes.
  - *This smart material-based system is still in the experiment stage but holds promise for the future.*

Recovering materials and re-processing methods…

- Differentiated bins and refuse collections facilitate easier sorting of waste for recycling.
- Materials such as aluminium recycle easily, but CFRP and other composites don’t.

Recovering materials and re-processing methods…

- Maintaining materials in a pure state is crucial and some are easily contaminated.
  - *One PVC bottle in a batch of 10,000 PET bottles can ruin the recycled PET.*
  - *Making recycled PET food-safe is difficult as it absorbs contaminants, so most PET containers are recycled as fibres.*

Recovering materials and re-processing methods…

- Polymer identification codes promote more effective and appropriate sorting and disposal.
- ***Materials recovery facilities (MRF)*** have to comply with a strict code of practice. Their manual systems are increasingly being supplemented by optical / digital automatic sorting systems, particularly ***Near Infra Red (NIR) technology***, to separate different types of waste.

Plastic Resin Identification Codes

1 PETE	2 HDPE	3 PVC	4 LDPE	5 PP	6 PS	7 OTHER
Polyethylene Terephthalate	**High-Density Polyethylene**	**Polyvinyl Chloride**	**Low-Density Polyethylene**	**Polypropylene**	**Polystyrene**	**Other**
Common products: soda & water bottles; cups, jars, trays, clamshells	Common products: milk jugs, detergent & shampoo bottles, flower pots, grocery bags	Common products: cleaning supply jugs, pool liners, twine, sheeting, automotive product bottles, sheeting	Common products: bread bags, paper towels & tissue overwrap, squeeze bottles, trash bags, six-pack rings	Common products: yogurt tubs, cups, juice bottles, straws, hangers, sand & shipping bags	Common products: to-go containers & flatware, hot cups, razors, CD cases, shipping cushion, cartons, trays	Common types & products: polycarbonate, nylon, ABS, acrylic, PLA; bottles, safety glasses, CDs, headlight lenses
Recycled products: clothing, carpet, clamshells, soda & water bottles	Recycled products: detergent bottles, flower pots, crates, pipe, decking	Recycled products: pipe, wall siding, binders, carpet backing, flooring	Recycled products: trash bags, plastic lumber, furniture, shipping envelopes, compost bins	Recycled products: paint cans, speed bumps, auto parts, food containers, hangers, plant pots, razor handles	Recycled products: picture frames, crown molding, rulers, flower pots, hangers, toys, tape dispensers	Recycled products: electronic housings, auto parts,
						

Recovering materials and re-processing methods…

- **Apple** has developed a ***robot*** named ***Daisy*** that can separate parts and components of up to 200 iPhones an hour and is capable of dealing with a wide range of different models.
- ***'Cradle to grave'*** means consideration of the impact of the whole life cycle of products. This includes *raw material extraction and production, manufacture, distribution, use and disposal*.

Apple Made a Robot - Meet Daisy…



https://www.youtube.com/watch?v=MC81peMLEfo

Recovering materials and re-processing methods…

- A ***'cradle to cradle'*** approach is even better, as this means that all materials in a product can be reused.
  - Accreditation has been introduced for ***'cradle to cradle'*** guarantee that materials remain separate for easy disassembly and return to the stock of raw material in pure form.
  - The ***'Think Chair'*** was one of the first certified products.

The *'Think Chair'*…

https://www.steelcase.com/eu-en/products/office-chairs/think/



Recovering materials and re-processing methods…

- It is alleged that dangerous, toxic processes for exported electronic waste are affecting the health of recycling workers, including children, who carry these out in places like Guiyu China.

Disposal to Landfill…

- Landfill sites are specially licensed locations such as disused quarries and land reclamation areas where non-recycled waste is buried in large quantities.
- They are generally convenient and safe places to deposit waste, but this depends on the installation of several safety systems, and they have a number of disadvantages.

Disposal to Landfill…

Disadvantages:

- Waste such as rotting food results in the production of methane and carbon dioxide and also requires deodorisation to reduce unpleasant smells.
- If sufficient care is taken it is possible for chemical to leak out and streams and ground water.

Disposal to Landfill…

Disadvantages:

- Although relatively cheap to build and run, monitoring them after they are full and capped is a potentially never-ending expense.
- It is likely that most of the embodied energy and materials contained in landfill will never be accesses.
- Scotland intends to completely ban biodegradable municipal waste (BMW) from landfill in 2021.

Energy Recovery…

- Local authorities are now starting to make more use of Recycling and Energy Recovery Facilities (RERF) to reduce their dependence on landfill sites.
- A minimum of ten per cent of general waste collected is extracted for recycling.
- A specialised combustion process is used to create steam and generate electricity for the national grid or district heating schemes.
- The remaining ash is recycled for use as aggregate in construction.

Now Test Yourself

1. *Explain two advantages and two disadvantages of using biopolymers for the manufacture of a specific product.*
2. *Explain three contrasting ways that energy costs could be minimised in the manufacture of products.*
3. *Suggest three ways in which a food manufacturer might be able to improve the energy efficiency of their product distribution.*

Now Test Yourself

4. *Use an example, other than those mentioned in this book, of a 'modular' design and explain how the designer has achieved this and what sustainability benefits may result.*
5. *Explain, using examples, why recycling polymer products can be problematic.*